An Adaptive IDS Model Based on Swarm Intelligence and Support Vector Machine

Surat Srinoy
Faculty of Science and Technology
Suan Dusit Rajabhat University, Bangkok, Thailand
Tel: +66-2-244-5225, Fax: +66-2-668-7136
E-mail: surat_sri@dusit.ac.th

Abstract— **Intrusion detection system looks for unusual or suspicious activity, such as patterns of network traffics that are likely indicators of unauthorized activity. New intrusion types, of which detection systems are unaware, are the most difficult to detect. The amount of available network audit data instances is usually large, human labeling is tedious, time-consuming, and expensive. In this paper we present support vector machine approach to data clustering. Support vector machine is used to initially create raw clusters and then these clusters are refined using Artificial Fuzzy Ants Clustering (AFAC). AFAC that has been developed as swarm intelligence techniques. The Algorithm uses ant colony optimization principle to find good partitions of the data. Certain unnecessary complications of the original algorithm are discussed and means of overcoming these complexities are proposed. We propose Artificial Fuzzy Ants Clustering (AFAC) in the second stage for refinement mean of overcoming these complexities are proposed. Our approach allows us to recognize not only known attacks but also to detect suspicious activity that may be the result of a new, unknown attack. The experimental results on Knowledge Discovery and Data Mining-(KDDCup 1999)** .

I. INTRODUCTION

Intrusion detection is a problem of great significance to protecting information systems security; especially in view of the internetworking is a crucial aspect of daily life around the world increasing incidents of cyber attacks on the critical infrastructures. It includes attempting to destabilize the network, gaining unauthorized access to files with privileges, or mishandling and misusing of software. The intrusion detection is to automatically scan network activity and detection attacks. As defined in [1], intrusion detection is "the process of monitoring the events occurring in a computer system or network and analyzing them for signs of intrusions. It is also defined as attempts to compromise the confidentiality, integrity, availability, or to bypass the security mechanisms of a computer or network".

There have been many techniques for modeling anomalous and normal behaviors for intrusion detection. The signature-based and supervised anomaly detections are widely deployed and commercially available. The signature-based detection extracts features from the network data. It detects intrusions by comparing the feature values to a set of attack signatures provided by human experts. However, it can only detect previously known intrusions with a signature. The signature database has to be manually revised for each new type of discovered attacks. On the other hand, the supervised anomaly detection trains models on labeled data (i.e., data pre-classified as an attack or not) and checks how well new data fit into the model. Obviously, it cannot be quickly adapted to new types of intrusion and do not have enough labeled data available. In general, a very large amount of network data needs to be handled and classified. Hence, it is impractical to classify them manually.

Intrusion detection techniques can be categorized in misuse detection and anomaly detection [2]. Misuse detection systems find intrusions by matching sample data to known intrusive pattern. Anomaly detection systems find intrusion by analyzing the deviation from normal activities profiles that are retrieved from historical data. Intrusion detection is a critical component of secure information systems. Many approaches have been proposed which include statistical [3], machine learning [4], data mining [5] and immunological inspired techniques [6]. Statistical analysis techniques are widely used in anomaly detection [7]. Compared with machine learning methods, statistical analysis techniques have an advantage that they can run in real time without offline learning and relearning from training data, but its detection performance is not good enough [8].

Support Vector Machine (SVM) are new universal machine learning methods based on statistical learning theory proposed by Vapnic and co-workers [9]. We construct a prototype of anomaly detection model using support vector machine is used to initially create raw clusters and then these clusters are refined using Artificial Fuzzy Ants Clustering (AFAC). AFAC that has been developed as swarm intelligence techniques. The Algorithm uses ant colony optimization principle to find good partitions of the data.

The rest of this paper is organized as follows. In section II, we discuss the related works; in section III describe a brief introduction to support vector machine and artificial fuzzy ant clustering in section IV; experimental design explains in section V, evaluates our intrusion detection model through experiments in section VI. Finally, section VII presents our conclusion, some discussion and future work.

II. RELATED WORKS

Authorized licensed use limited to: Universidad Nacional de Ingeniería. Downloaded on October 19,2026 at 03:43:58 UTC from IEEE Xplore. Restrictions apply.

In a classification problem, the number of features can be quite large, many of which can be irrelevant or redundant. Since the amount of audit data that an IDS needs to examine is very large even for a small network, classification by hand is impossible. Feature reduction and feature selection improves classification by searching for the subset of features, which best classifies the training data. Some of the important features an intrusion detection system should possess include refer in Srilatha et al. [10].

Most intrusion occurs via network using the network protocols to attack their targets. Twycross [11] proposed a new paradigm in immunology, Danger Theory, to be applied in developing an intrusion detection system. Alves et al. [12] presents a classification-rule discovery algorithm integrating artificial immune systems (AIS) and fuzzy systems. For example, during a certain intrusion, a hacker follows fixed steps to achieve his intention, first sets up a connection between a source IP address to a target IP, and sends data to attack the target [13]. Generally, there are four categories of attacks [11]. They are: 1) DoS (denial-of-service), for example ping-of-death, teardrop, smurf, SYN flood, and the like. 2) R2L : unauthorized access from a remote machine, for example guessing password, 3) U2R : unauthorized access to local super user (root) privileges, for example, various "buffer overflow" attacks, 4) PROBING: surveillance and other probing, for example, port-scan, ping-sweep, etc. Some of the attacks (such as DoS, and PROBING) may use hundreds of network packets or connections, while on the other hand attacks like U2R and R2L typically use only one or a few connections[12].

III. SUPPORT VECTOR MACHINE

Support Vector Machines have been proposed as a novel technique for intrusion detection. SVM maps input (real-valued) feature vectors into higher dimensional feature space through some nonlinear mapping. SVMs are powerful tools for providing solutions to classification problems. These are developed on the principle of structural risk minimization. Structural risk minimization seeks to find a hypothesis for which one can find the lowest probability of error. The structural risk minimization can be achieved by finding the hyper plane with maximum separable margin for the data [14].

To automatically separate IDS data into normal or anomalous distributions. The intrusion detection model based on SVM (support vector machine) is mainly classified into three types. The first type [15] divides data into normal data and attack data using characteristics of the binary classifier SVM, and the second type [16] implements an anomaly detection model using one-class SVM. Finally, the third type [17] establishes multi-class SVM at the form of combining many binary classifiers, namely, SVMs and divides data into normal data and four types of attack data [18].

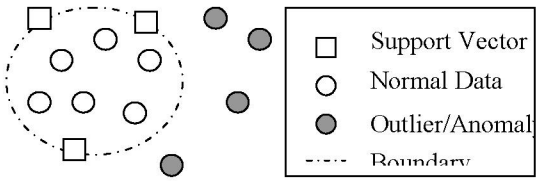


Figure 1. Sample SVM

In Fig. 1 normal (within bounds) and Anomalous (outliers). Those data points that define the boundary between the normal and the anomalous are called support vectors, thus Support Vector Machines. These "define" normalcy.

Computing the hyper plane to separate the data points, i.e. training a SVM, leads to a quadratic optimization problem. SVM uses a feature called a kernel to solve this problem. A kernel transforms linear algorithms into nonlinear ones via a map into feature spaces. SVMs classify data by using these support vectors, which are members of the set of training inputs that outline a hyper plane in feature space [19].

The kernels very useful in finding and generalizing the principles that distinguish normal data from attacks. It is in this flexibility that the use of SVM's and kernels can greatly increase the IDS success rate at detecting new attacks.

In [20] explained kinds of learning machines as follows.

Two-class classification. SVM learn linear decision rules described by a weighted vector and a threshold. The idea of structural risk minimization is to find a hypothesis for which one can guarantee the lowest probability of error. Geometrically, it can find two parts representing the two classes with a hyper-plane with normal and distance from the origin as depicted in Figure 2.

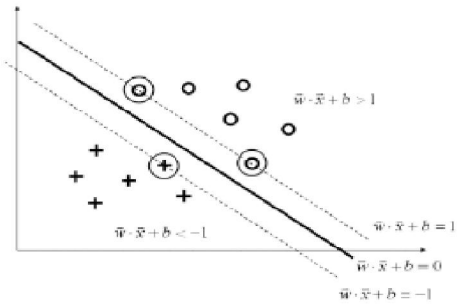


Figure 2. Optimal separating hyper plane is the one that separates the data

$$D(\vec{x}) = sign\{\vec{\omega}.\vec{x} + b\} = \begin{cases} +1, if\ \vec{\omega}.\vec{x} + b > 0 \\ -1, \quad \text{else} \end{cases} \qquad (1)$$

Support such separating hyper-planes exist. It defines the margin of a separating hyper-plane as the minimum distance between all input vectors and the hyper-plane.

Multi-class Classification. SVM were originally designed for two-class classification, commonly called as binary classification. Currently there are two types of approaches for

Authorized licensed use limited to: Universidad Nacional de Ingeniería. Downloaded on October 19,2026 at 03:43:58 UTC from IEEE Xplore. Restrictions apply.

multi-class SV<, one-against-all [21] and one-against-one approaches [22]. One is by constructing and combining several binary classifiers while the other is by directly considering all data in one optimization formulation [23].

IV. ARTIFICIAL FUZZY ANT CLUSTERING

The ants cluster the objects to form heaps. A heap is spatially located in a single cell. Consider a heap H with n_H objects, then can define the following parameters [24]:

- The maximum distance between two s-dimensional objects in the heap

$$Dst_{\max}(H) = \max_{X_i, X_j \in H} D(Xi, Xj) \qquad (1)$$

Where *Dst* is euclidean the distance between the objects

- The center of all the objects in the heap

$$C_{center}(H) = \frac{1}{n_H} \sum_{Ci \in H} C_i \qquad (2)$$

- The most dissimilar object in the heap $O_{dissim}(H)$. It is the object which is the farthest from the center of the heap.
- The mean distance between the objects of H and the center of the heap

$$Dst_{mean}(H) = \frac{1}{n_H} \sum_{O_i \in H} Dst(O_i, C_{center}(H)) \quad (3)$$

The main ant based clustering algorithm is presented in figure 3 [25].

1. Randomly place the ants on the board. Randomly place objects on the board at most one per cell
2. Repeat
3. For each ant Do
   a. Move the ant
   b. If the ant doest not carry any object then if there is an object in the 8 neighboring cells of the ant, the ant possibly picks up the objects,
   c. Else the ant possibly drops a carried object, by looking at eht 8 neighboring around it
4. Until stopping criteria

Figure 3. The basic ant based algorithm [25]

Initially the ants are scattered randomly on the 2D board. The ant moves on the board and possibly picks an object or drops an object. The movement of the ant is not completely random. Initially the ant picks a direction randomly then the ant continues in the same direction with a probability $P_{direction}$, otherwise it generates a new random direction. On reaching the new location on the board the ant may possibly pick up an object or drop an object, if it is carrying one. The heuristics and the exact mechanism for picking up or dropping an object are explained below. The stopping criterion for the ants, here, is the upper limit on the number of times through the repeat loop.

A. *Picking up an object*

When the ant is not carrying any object, it looks for possible objects to pick up by looking at the eight neighboring cells around its current position. If an object or heap is found then the ant possibly picks up an object. The heuristic for picking up an object depends on the number of objects in the heap. Three cases are considered: only one object, a heap of two objects and a heap of more than two objects. If a single object is present then the ant has a fixed probability of picking it up. If there is a heap

of two objects then with a probability $P_{destroy}$ the ant destroys the heap by picking a random object from the heap. In the third case the ant picks up the most dissimilar object from the heap if the dissimilarity is above a given threshold T_{remove}.

The algorithm for picking up an object is given in figure 4 [25].

1. Mark the 8 neighboring cells around the ant as *"unexplored"*
2. Repeat
   2.1 Consider the next unexplored cell around the ant
   2.2 If the cell is not empty then
      2.2.1 If the cell contains a single object *X*, then the object *X* is picked up with a probability *Pload*, else
      2.2.2 If the cell contains a heap of two objects, then the heap is destroyed by picking up a random object with a probability *Pdestroy*
      else
      2.2.3 If the cell contains a heap H of more than 2 objects, then the most dissimilar object, *Odissim*(*H*), of H is removed only if
      $$\frac{Dst(O_{dissim}(H), C_{center}(H))}{Dst_{mean}(H)} > T_{remove}$$
   2.3 Label the cell as *"explored"*
3. Until all the neighboring cells have been explored or one object has been picked

Figure 4. Algorithm to pick up an object [25]

B. *Dropping an object*

When the ant is carrying an object, then it examines the 8 cells surrounding its current location. Three cases are considered: the cell is empty, the cell contains one object only, and the cell contains a heap. In the first case the ant has a constant probability of dropping the object. In the second case a heap is created if the carried object is sufficiently similar to the one already in the cell. In the third case the ant will add its object to the heap if the object is closer to H's center than the most dissimilar object of H. The algorithm for dropping the object is given in Figure 5 [25].

C. *The Second Stage*

In the second stage we consider the heaps formed by the first stage and move the entire heap on the 2D board. The algorithm for picking up a heap is the same as that for the objects. Ants will pick the heap with the same probability P_{load}. Ants drop a heap H_1 onto another heap H_2 provided that:[24]

Authorized licensed use limited to: Universidad Nacional de Ingeniería. Downloaded on October 19,2026 at 03:43:58 UTC from IEEE Xplore. Restrictions apply.

$$\frac{Dst(O_{center}(H_1), O_{center}(H_2))}{Dst_{\max}} < T_{createforheap} \quad (4)$$

When two heaps *H1* and *H2* are merged, they form a single heap *H3*. They cannot be separated. The number of heap goes on decreasing as the number of iterations increases [24].

1. Mark the 8 neighboring cells around the ant as *"unexplored"*
2. Repeat
   2.1 Consider the next unexplored cell around the ant
      2.1.1 If the cell is empty then the object carried by the ant, *X*, is dropped with a probability *Pdrop* else
      2.1.2 If the cell contains a single object *X'* then a heap of two objects is created by dropping *X* on *X'* only if
      $$\frac{Dst(X, X')}{Dst_{\max}} < T_{create}$$ else
      2.1.3 If the cell contains a heap *H* then *X* is dropped on *H* only if
      $$D(X, C_{center}(H)) < D(O_{dissim}(H), C_{center}(H))$$
   2.2 Label *c* as *"explored"*
3. Until all the neighboring cells have been explored or the carried object has been dropped.

Figure 5. Algorithm for dropping an object [25]

V. EXPERIMENTAL SETUP AND RESULT

In this experiment, we use a standard dataset the raw data used by the KDD Cup 1999 intrusion detection contest [26]. This database includes a wide variety of intrusions simulated in a military network environment that is a common benchmark for evaluation of intrusion detection techniques. In general, the distribution of attacks is dominated by probes and denial-of-service attacks; the most interesting and dangerous attacks, such as compromises, are grossly under-represented [27]. The data set has 41 attributes for each connection record plus one class label. There are 24 attack types, but we treat all of them as an attack group. A data set of size N is processed. The nominal attributes are converted into linear discrete values (integers). After eliminating labels, the data set is described as a matrix *X*, which has *N* rows and *m*=41 columns (attributes). There are m_d=8 discrete-value attributes and m_c = 33 continuous-value attributes.

We ran our experiments on a system with a 1.6 GHz Pentium M processor and 512 MB DDR RAM running Windows XP. All the preprocessing was done using MATLAB®. MATLAB's Toolbox [28]. In practice, the number of classes is not always known beforehand. There is no general theoretical solution to finding the optimal number of clusters for any given data set. We choose k = 5 for the study. We will compare five classifiers which have been also used in detecting these four types of attacks.

D. Data preprocessing

A considerable amount of data-preprocessing had to be undertaken before we could do any of our modeling experiments. It was necessary to ensure though, that the reduced dataset was as representative of the original set as possible. The test dataset that previously began with more than 300,000 records was reduced to approximately 18,216 records. Table 1 shows the dataset after balanced among category for attack distribution over modified the normal and other attack categories. Preprocessing consisted of two steps. The first step involved mapping symbolic-valued attributes to numeric-valued attributes and the second step implemented non-zero numerical features. We reduce the dimensionality of this data set from 41 to 10 attributes are *duration, service, src_bytes, dst_byte, count, srv_count, serror_rate, dst_host_srv_count, dst_host_diff_srv_rate, and dst_host_same_src_port_rate.*

Table 1. Dataset for attack distribution

Attack Category	% Occurrence	Number of records
normal	31.64	5,763
probe	11.88	2,164
DoS	19.38	3,530
U2R	0.38	70
R2L	36.72	6,689
Summary	100	18,216

E. Feature Selection

Feature selection techniques aim at reducing the number of unnecessary features in classification rules. Support Vector Machine has been used to define the necessity of features.

Feature selection is an optimization process in which one tries to find the best feature subset, from the fixed set of the original features, according to a given processing goal and a feature selection criterion. A pattern's features, from the point of view of processing goal and type, may be irrelevant (having no effect on processing performance) or relevant (having an impact on processing performance). Features can be redundant (correlated, dependent) [30]. When we process volumes of data, it is necessary to reduce the large number of features to a smaller set of features. There are 42 fields in each data record and it is hard to determine which fields are useful or which fields are trivial. Jin et al [33] suggest correlation coefficients between fields by using SPSS. They propose that if the correlation coefficients of fields *i* and *j*, *R(i,j)*, is larger than 0.8, then there is a strong correlation between fields *i* and *j*, and will select either one of them to represent these two fields. Support vector machine allow us to determine (for a discrete attribute data set) a set called a core, containing strongly relevant features, and reducts, containing core plus additional weakly relevant features, such that each reduct is satisfactory to determine concepts in the data set. Based on a set of reducts for a data set some criteria for feature selection can be formed, for example a selecting

Authorized licensed use limited to: Universidad Nacional de Ingeniería. Downloaded on October 19,2026 at 03:43:58 UTC from IEEE Xplore. Restrictions apply.

feature from a reduct containing the minimal set of attributes [30].

F. Performance Measure

Standard measures for evaluating IDSs include *detection rate*, *false alarm rate*, *trade-off between detection rate and false alarm rate* [31], *performance* (Processing speed + propagation + reaction), and *Fault Tolerance* (resistance to attacks, recovery, and subversion). Detection rate is computed as the ratio between the number of correctly detected attacks and the total number of attacks, while false alarm (false positive) rate is computed as the ratio between the numbers of normal connections that are incorrectly misclassified as attacks [32]. These are good indicators of performance, since they measure what percentage of intrusions the system is able to detect and how many incorrect classifications are made in the process.

Anomaly detection amounts to training models for normal traffic behavior and then classifying as intrusions any network behavior that significantly deviates from the known normal patterns and to construct a set of clusters based on training data to classify test data instances.

VI. CONCLUSIONS AND FUTURE WORK

In this paper we apply support vector machine methods to intrusion detection to avoid a hard definition between normal class and certain intrusion class and could be considered to be in more than one category (or from another point of view it allows representation of overlapping categories).We introduce the current status of intrusion detection systems (IDS) and artificial fuzzy ants clustering based feature selection heuristics , and present some possible data mining based ways for solving problems. AFAC based methods with data reduction for network security are discussed. Intrusion detection model is a composition model that needs various theories and techniques. One or two models can hardly offer satisfying results. We plan to apply other theories and techniques in intrusion detection in our future work.

ACKNOWLEDGMENT

This work is supported by Suan Dusit Rajabhat University via lecturer research project grant. Thank you Parag M. Kanade and Lawrence O. Hall in [24] for source of algorithm in this paper.

REFERENCES

[1] D.S Bauer, M.E Koblentz,. "NIDX- an expert system for real-time network intrusion detection", *Proceedings of the Computer Networking Symposium*, 1988. pp. 98-106.

[2] Herv Debar, Marc Dacier, and Andreas Wespi, "Towards a Taxonomy of Intrusion Detection Systems", IBM Technical Paper, Computer Networks, Vol.39, Issue 9, pp. 805-822, April 1999.

[3] R. Bace and P. Mell, "Intrusion Detection Systems", NIST Special Publication on Intrusion Detection System, 31 November 2001.

[4] A.Sundaram, "An introduction to intrusion detection, Crossroads": The ACM student magazine, 2(4), April 1996.

[5] D. Denning, "An intrusion-detection model", In IEEE computer society symposium on research in security and privacy, 1986, pp. 118-131.

[6] T.Lane, "Machine Learning techniques for the computer Security", PhD thesis, Purdue University, 2000.

[7] Madori Asaka, Takefumi Onabuta, Tadashi Inoue, Shunji Okazawa, and Shigeki Goto, "A new intrusion detection method based on discriminant analysis", Oakland, pp.130-143, May 2001.

[8] Ming Tang, Song-Can Chen, Yi Zhuang and Jia Liu, "Using Statistical Analysis and Support Vector Machine Classification to Detection Complicated Attacks", Proc. Of the Third International Conference on Machine Learning and Cybernetics IEEE Computer Society, Shanghai, China, August 26-29, 2004.

[9] N. Cristianini and J. Shawe-Taylor, "An Introduction to Support Vector Machines," Cambridge University Press, Cambridge, UK, 2000.

[10] S. Chebrolu, A. Abraham, J. P. Thomas, *Feature Deduction and Ensemble Design of Intrusion Detection Systems*, Computer & Security, 2004.

[11] J. Twycross , Immune Systems, "Danger Theory and Intrusion Detection", *presented at the AISB 2004 Symposium on Immune System and Cognition*, Leeds, U.K., March 2004.

[12] R.T. Alves, M.R.B.S. Delgado, H.S. Lopes, A.A. Freitas, "An artificial immune system for fuzzy-rule induction in data mining", *Lecture Notes in Computer Science, Berlin: Springer-Verlag*,Vol.3242, 2004, pp.1011-1020.

[13] W. Lee and S. Stolfo, Data Mining Approaches for Intrusion Detection, *Proceedings of the 7th USENIX security symposium*, 1998.

[14] V.N.Vapnik, *"The Nature of Statistical Learning Theory"*, Springer, 1995.

[15] W.H. Chen, S.H. Hsu, and H.P. Shen, "Application of SVM and ANN for intrusion, Computers & Operations Research", *ELSEVIER*, Vol. 32, No.10, 2005, pp. 2617-2634.

[16] K.L. Li, H.K. Huang, S.F. Tian, and W. Xu, "Improving one-class SVM for anomaly detection, International Conference on Machine Learning and Cybernetics", Vol.5, 2003, pp. 3077-3081.

[17] T. Ambwani, "Multi class support vector machine implementation to intrusion detection", Proceedings of the International Joint Conference on Neural Networks, Vol. 3, 2003, pp. 2300-2305.

[18] H. Lee, J. Song, D. Park, "Intrusion Detection System Based on Multi-class SVM", RSFDGrC, Vol. 2, 2005, pp. 511-519.

[19] A. Abraham, R.Jain, "Soft Computing Models for Network Intrusion Detection Systems CoRRcs", CR/0405046, 2004.

[20] J. C.Christopher Burges, "A Tutorial on Support Vector Machines for Pattern Recognition, *Data Mining and Knowledge Discovery*", 1998.

[21] B. Schoelkopf, C. Burges, and V. Vapnik., "Extracting support data for a given task". In U.M. Fayyad, and R. Uthurusamy, editors, Proceedings, *First International Conference on Knowledge Discovery & Data Mining*, AAAI Press, MenloPark, CA ,1995.

[22] S. Knerrr, L. Personnaz, and G. Dreyfus, "Single-layer Learning Revisited: a Stepwise Procedure for Building and Training a Neural Network", In J. Fogelman, editor, Neurocomputing: Algorithms, *Architectures and Applications, Springer-Verlag*, 1990.

[23] D. S. Kim, J. S. Park, "*Network-Based Intrusion Detection with Support Vector Machines"*, ICOIN, 2003, pp. 747-756.

Authorized licensed use limited to: Universidad Nacional de Ingeniería. Downloaded on October 19,2026 at 03:43:58 UTC from IEEE Xplore. Restrictions apply.

[24] P. M. Kanade and L. O. Hall. Fuzzy Ants as a Clustering Concept. [22]nd international conference of the North American fuzzy information processing society NAFIPS, p. 227-232

[25] Monmarche N., Silmane M., and Venturini G., "AntClass: discovery of clusters in numeric data by an hybridization of an ant colony with k-means algorithm", *Internal Report no. 213,* Laboratoire d'Informatique de l'Universite de January 1999.

[26] KDD data set, 1999; http://kdd.ics.uci.edu/databases/kddcup99/kddcup99.html

[27] P. Laskov, K. Rieck, C. Schäfer, K.R. Müller, "Visualization of anomaly detection using prediction sensitivity", *Proc.of Sicherheit,* April 2005, 197- 208.

[28] Math Works, *Statistical Toolbox for User's Guide, Math Works*, 2001.

[29] K. Cios, W. Pedrycz, R. Swiniarski, *Data Mining-Methods for Knowledge discovery, Kluwer Academic Publishers*, 1998.

[30] W. Chimphlee, Abdul Hanan Abdullah, Mohd Noor Md Sap and S. Chimphlee, "Unsupervised Anomaly Detection with Unlabeled Data Using Clustering", *Proc. Int. Conf. on ICT-Mercu Buana ICT2005*. 42-49.

[31] A. Lazarevic, A. Ozgur, L. Ertoz, J. Srivastava, and V. Kumar, A comparative study of anomaly detection schemes in network intrusion detection. *In SIAM International Conference on Data Mining*, 2003.

[32] T. Wakaki, H. Itakura, and M.Tamura, Rough Set-Aided Feature Selection for Automatic Web-Page Classification, *Proc. of the IEEE/WIC/ACM International Conference on Web Intelligence (WI'04)*

[33] H. Jin, J. Sun, H. Chen, and Z. Han, A Fuzzy Data Mining Based Intrusion Detection System, *Proc. of 10thInternational Workshop on future Trends in Distributed Computing Systems (FTDCS04) IEEE Computer Society*, Suzhou, China, May 26-28, 2004, 191-197.

Authorized licensed use limited to: Universidad Nacional de Ingeniería. Downloaded on October 19,2026 at 03:43:58 UTC from IEEE Xplore. Restrictions apply.